(i)

$$u_{n+1} = \tfrac{1}{2}\left(u_{n+2} + u_n\right) \implies 2u_{n+1} = u_{n+2} + u_n$$

$$\implies u_{n+2} - u_{n+1} = u_{n+1} - u_n$$

$$\therefore \boxed{\text{the difference between successive terms is constant}}$$

Let $k = u_2 - u_1$,

$$u_n = u_1 + (n-1)k$$

which is a polynomial in n of degree 1 if $k \neq 0$ and of degree 0 if $k = 0$

$$\therefore \boxed{u_n \text{ has degree at most 1}}$$

(ii) Substituting $v_n = t_n + pn^2$ into $v_{n+1} = \tfrac{1}{2}\left(v_{n+2} + v_n\right) - p$:

$$t_{n+1} + p(n+1)^2 = \tfrac{1}{2}\left(t_{n+2} + p(n+2)^2 + t_n + pn^2\right) - p$$

$$\tfrac{1}{2}\left(p(n+2)^2 + pn^2\right) - p = p\left(n^2 + 2n + 2\right) - p = p(n+1)^2$$

so the terms in p cancel

$$\implies t_{n+1} = \tfrac{1}{2}\left(t_{n+2} + t_n\right)$$

By part (i), t_n has degree at most 1, and $p \neq 0$

$$\therefore \boxed{v_n \text{ has degree 2}}$$

Write $v_n = pn^2 + qn + r$

$$v_1 = p + q + r = 0, \qquad v_2 = 4p + 2q + r = 0$$

$$3p + q = 0 \implies q = -3p, \qquad r = -p - q = 2p$$

$$\therefore \boxed{v_n = p\left(n^2 - 3n + 2\right) = p(n-1)(n-2)}$$

(iii) Substituting $w_n = t_n + kn^3$ into $w_{n+1} = \tfrac{1}{2}\left(w_{n+2} + w_n\right) - an - b$:

$$t_{n+1} + k(n+1)^3 = \tfrac{1}{2}\left(t_{n+2} + k(n+2)^3 + t_n + kn^3\right) - an - b$$

$$\tfrac{1}{2}\left(k(n+2)^3 + kn^3\right) = k\left(n^3 + 3n^2 + 6n + 4\right), \qquad k(n+1)^3 = k\left(n^3 + 3n^2 + 3n + 1\right)$$

so the terms in kn^3 and kn^2 cancel, leaving

$$t_{n+1} = \tfrac{1}{2}\left(t_{n+2} + t_n\right) + (3k - a)n + (3k - b)$$

Choosing $k = \frac{1}{3}a$:

$$t_{n+1} = \tfrac{1}{2}\left(t_{n+2} + t_n\right) - (b - a)$$

$b \neq a$: this is part (ii) with $p = b - a$, so t_n has degree 2 with leading coefficient $b - a$
$b = a$: this is part (i), so t_n has degree at most 1, and the coefficient of n^2 is $0 = b - a$
In both cases t_n has degree at most 2 and its coefficient of n^2 is $b - a$, and $a \neq 0$

$$\therefore \boxed{w_n \text{ has degree } 3}$$

Write $w_n = \frac{1}{3}an^3 + (b - a)n^2 + qn + r$

$$w_1 = \tfrac{1}{3}a + (b - a) + q + r = 0$$

$$w_2 = \tfrac{8}{3}a + 4(b - a) + 2q + r = 0$$

Subtracting: $\frac{7}{3}a + 3(b - a) + q = 0$

$$q = 3a - \tfrac{7}{3}a - 3b = \tfrac{2}{3}a - 3b$$

$$r = -\tfrac{1}{3}a - (b - a) - q = \tfrac{2}{3}a - b - \tfrac{2}{3}a + 3b = 2b$$

$$\therefore \boxed{w_n = \tfrac{1}{3}an^3 + (b - a)n^2 + \left(\tfrac{2}{3}a - 3b\right)n + 2b}$$

$\because w_1 = w_2 = 0$

$$\therefore w_n = \tfrac{1}{3}(n - 1)(n - 2)(an + 3b)$$

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